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Present:

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Start of meeting

1 Meeting Discussion

1.1 Signal Windowing & Validation Strategy

- **Overlapping Windows:** Idea of implementing sliding windows with an overlap on the historical time-series data to improve feature density.
- 5 • **Data Splitting:** Ensure a strict and proper separation between the Training set and the Test set to prevent data leakage, especially when dealing with overlapping windows.

1.2 Feature Selection & Defense Framework

- **Scientific Justification:** It is highly critical to rigorously justify the selection of the extracted features. Clear, measurable criteria must be established to explain why certain features were selected and others
10 discarded, as this will be a major focus of challenge during the final thesis defense.
- **Automated Feature Relevance:** Investigate specialized time-series libraries capable of extracting, testing, and automatically filtering out irrelevant features (specifically `tsfresh`) to establish a mathematically sound selection baseline.

1.3 Hyperparameter Optimization

- 15 • Explore automated strategies for hyperparameter tuning. Evaluation should be conducted regarding whether to use a standard Grid Search approach or a more advanced optimization framework via `Optuna` or others framework (TBD).

1.4 Milestone Deliverables

- 20 • A formal project report update must be pushed to the repository at the end of each related,important milestone.

2 Action Items & Short-Term Roadmap

- **Feature Engineering:** Ensure the `tsfresh` feature extraction pipeline is fully operational and optimized on the Disco computing environment.
- 25 • **Model Implementation:** Deploy functional baselines for both TFT (Temporal Fusion Transformer) and NBEATSx. Establish a definitive comparison framework based on their architectural differences.
- **Hyperparameter Tuning:** Finalize the choice and implementation of the optimization framework (e.g., Optuna).
- **Documentation:** Parallely document all methodology, feature choices, and model architectures. Complete drafting by **08.06.2026**.

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Reformulated with Gemini